

Resonant Medium Theory — Symbols Table (v1.1)

Core Cosmology

Symbol	Meaning	Units
$a(t)$	Scale factor	—
$H(t) = da/dt / a$	Hubble parameter	1/s or km/s/Mpc
H_0	Hubble constant today	km/s/Mpc
z	Redshift	—
Ω_m	Matter density fraction today	—
Ω_{Λ}	Dark energy fraction (LambdaCDM)	—
$\rho(z)$	Energy density	kg/m ³ or J/m ³

Resonant Medium Variables

Symbol	Meaning	Units
τ	Memory depth parameter (lag in e-folds)	dimensionless
κ	Feedback polarity (restoring or destabilizing sign)	dimensionless
ω_0	Characteristic resonant frequency (log-scale)	rad per e-fold
A	Resonant mode amplitude	—
ϕ	Phase offset of mode	radians
$K(\Delta \ln a)$	Memory kernel	—
ϵ_E	Energy recycling efficiency to dark energy	—
ϵ_M	Mass recycling efficiency to dark matter	—

Cosmic Recycling Quantities

Symbol	Meaning	Units
$\rho_{BH}(z)$	Black hole mass density	solar masses per cubic megaparsec
$S_{BH}(z)$	Black hole source rate (accretion + mergers)	solar masses per year per cubic megaparsec
$\rho_{DE}(z)$	Recycled dark energy density	J/m ³

Symbol	Meaning	Units
$\rho_{\text{DM}}(z)$	Recycled dark matter density	kg/m^3
t_{E}	Dark energy recycling timescale	Gyr
t_{M}	Dark matter recycling timescale	Gyr

Observables and Data

Symbol	Meaning
$w(z)$	Dark energy equation of state
$f_{\text{sigma8}}(z)$	Structure growth rate observable
$C_{\text{I_Tg}}$	ISW temperature–galaxy cross spectrum
$C_{\text{I_gkappa}}$	Galaxy–lensing cross spectrum
A_{L}	CMB lensing amplitude
$\alpha_{\text{iso}}(z)$	Isotropic BAO scale
χ^2	Fit statistic
WWI	Weighted Win Index (model performance)

Constants

Symbol	Meaning	Value / Units
c	Speed of light	299,792,458 m/s
g	Gravitational constant	$6.67430\text{e-}11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
h	Planck constant	$6.62607015\text{e-}34 \text{ J}\cdot\text{s}$
$\hbar$	Reduced Planck constant	$1.054571817\text{e-}34 \text{ J}\cdot\text{s}$
H_0	Hubble constant today	$\sim 70 \text{ km/s/Mpc}$ (exact value from dataset)
σ_8	RMS matter fluctuation (8 Mpc/h)	~ 0.8 (dataset dependent)

Domains and Spaces

Symbol	Domain	Description
t	Time domain	cosmic time evolution
$\ln a$	Log expansion domain	used for resonance and memory analysis

Symbol	Domain	Description
$a(t)$	Scale factor domain	FRW cosmology base
ω	Frequency space	derived via Fourier over $\ln a$

Version

Resonant Medium Symbols Table — Version 1.2 Maintained by: Resonant Medium Research Program

This table defines the standard notation for all Resonant Medium work. Existing definitions will not change; new symbols may be added as needed. the standard notation for all Resonant Medium work. Existing definitions will not change; new symbols may be added as needed.